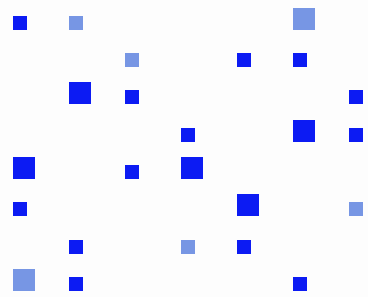


STORM DEATHS FELL PER RESIDENT.

THE EXPLANATION IS LESS SIMPLE.

THE CLAIM / AND ITS LIMIT

In the NWS historical series, U.S. tornado, flood, and hurricane deaths declined only modestly in raw counts from the 1940s to 2020–25. Relative to population, the decline is much larger. That is consistent with adaptation, but it does not identify which investments worked or how much risk remains.



1940s / 2020–25 (SIX YEARS)

1.88 / 0.68	262 / 230	140m / 336m
DEATHS PER MILLION RESIDENTS / YEAR	AVERAGE ANNUAL DEATHS / THREE CATEGORIES	AVERAGE U.S. POPULATION

FIGURE 1 / AVERAGE ANNUAL DEATHS BY NWS CATEGORY

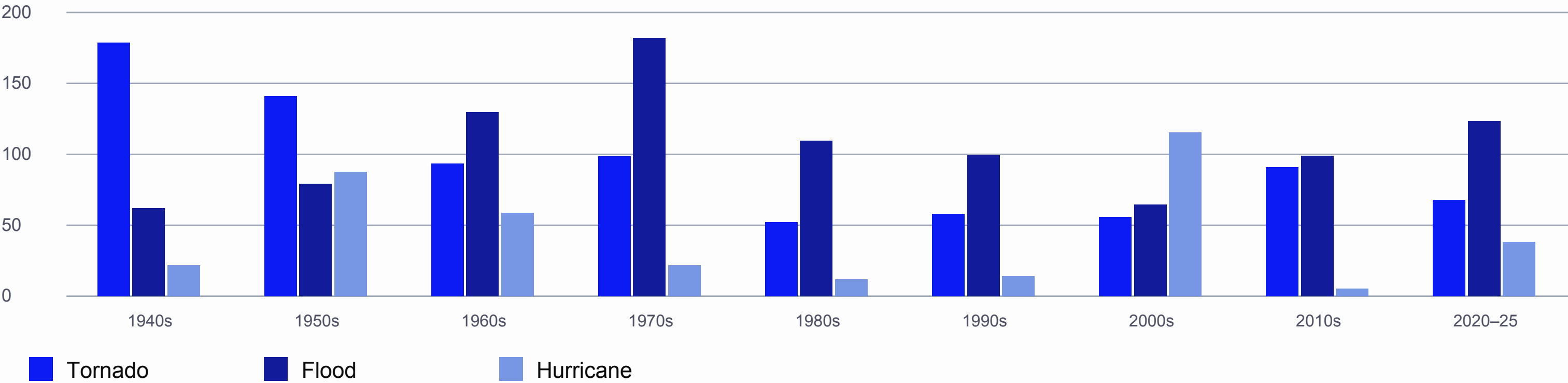
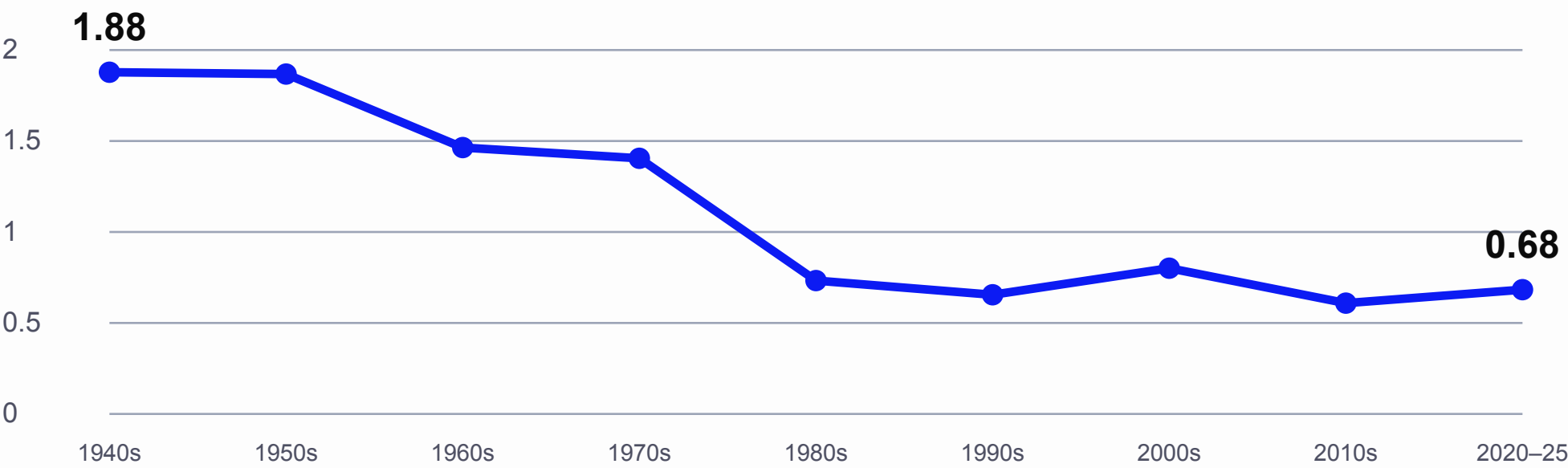


FIGURE 2 / COMBINED DEATHS PER MILLION U.S. RESIDENTS



READ THE DENOMINATOR

The per-resident index is about 64% lower at the endpoint. Flood deaths rose in raw counts, tornado deaths fell, and hurricane counts swing with major events. A national population denominator does not measure who actually faced a storm.

FLOOD MIGRATION

A Census study of 1999–2023 finds growth in blocks classified as flood-prone, but the parcel-level picture is steadier.

Flood-prone blocks growth

Exposed parcels 14.2–14.4%

TEMPERATURE CONTEXT

Burke et al. estimate that ambient temperature accounts for 5–12% of deaths in the countries studied, with cold exceeding heat. That modeled attribution is a different measure from NWS incident fatalities.

ECONOMICS

Better warnings, buildings, and response capacity may help explain the decline, but this series cannot isolate their effects. Compare the full cost of the next protective investment with expected avoided harm; use local exposure, not one national trend.

METHODS AND LIMITS

Annual NWS tornado + flood + hurricane deaths were divided by each year’s Census July 1 population (millions), then averaged within each period. 2020–25 has six years; 2025 NWS figures are preliminary. NWS reporting geography and Census population coverage do not align perfectly. This is a descriptive national index, not a causal or individual storm-risk measure. Heat is analyzed separately on the webpage because incident reports and modeled excess mortality answer different questions.



Sources: NWS, 81-Year Hazard Fatalities (1940–2025); U.S. Census Bureau, annual population estimates; Qin & Voorheis, Census WP 26-37; CDC, 1995 Chicago heat wave report; Burke et al., NBER WP 34313. Full methods, source links, and accessible data table: felix-ab.github.io/econ238-portfolio/week-four/